

PHYSICIAN & DEPARTMENT SQL

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Question :

Perform SQL operations on Physician and Department tables using INNER JOIN, WHERE, LIKE, NOT LIKE, IN, NOT IN, EXISTS, GROUP BY, HAVING, ORDER BY, subqueries, COUNT(), COUNT(DISTINCT), MIN(), and MAX() to retrieve and analyse physician and department information.

Step 1 : Create Physician Database and Table

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Displays a list of schemas including aggregate_bank_db, bank_management_db, bookflow_db, joins_operations_db, std_marks, and sys.
- Query Editor:** Contains the following SQL script:

```
1 CREATE DATABASE physician_department_db;
2 USE physician_department_db;
3 CREATE TABLE physician (
4     employeeid INT PRIMARY KEY,
5     name VARCHAR(100),
6     position VARCHAR(100),
7     ssn VARCHAR(20)
8 );
```
- Output:** Shows the execution results of the SQL script:

#	Time	Action	Message
1	19:45:03	CREATE DATABASE physician_department_db	1 row(s) affected
2	19:45:03	USE physician_department_db	0 row(s) affected
3	19:45:03	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100), ssn VARCHAR(20))	0 row(s) affected

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

Information

No object selected

SQL File 2*

Limit to 1000 rows

```

10 -- Create Physician table
11 CREATE TABLE physician (
12     employeeid INT PRIMARY KEY,
13     name VARCHAR(100),
14     position VARCHAR(100),
15     ssn VARCHAR(20)
16 );
17
18 INSERT INTO physician VALUES
19 (1, 'Geethika', 'Staff Internist', '111111111'),
20 (2, 'Siri', 'Attending Physician', '222222222'),
21 (3, 'Shailu', 'Surgical Attending Physician', '333333333'),
22 (4, 'Jhansi', 'Senior Attending Physician', '444444444'),
23 (5, 'Aishu', 'Head Chief of Medicine', '555555555'),
24 (6, 'Yamini', 'Surgical Attending Physician', '666666666'),
25 (7, 'Bhavitha', 'Surgical Attending Physician', '777777777'),
26 (8, 'Naksh', 'MD Resident', '888888888'),
27 (9, 'Mahija', 'Attending Psychiatrist', '999999999');
28

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duplicates	Warnings
6	19:55:28	CREATE DATABASE physician_department_db	1 row(s) affected	0	0
7	19:55:28	USE physician_department_db	0 row(s) affected	0	0
8	19:55:28	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR...	0 row(s) affected	0	0
9	19:55:28	INSERT INTO physician VALUES (1, 'Geethika', 'Staff Internist', '111111111), (2, 'S...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0	0

Object Info Session

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

Information

No object selected

SQL File 2*

Limit to 1000 rows

```

17
18 INSERT INTO physician VALUES
19 (1, 'Geethika', 'Staff Internist', '111111111'),
20 (2, 'Siri', 'Attending Physician', '222222222'),
21 (3, 'Shailu', 'Surgical Attending Physician', '333333333'),
22 (4, 'Jhansi', 'Senior Attending Physician', '444444444'),
23 (5, 'Aishu', 'Head Chief of Medicine', '555555555'),
24 (6, 'Yamini', 'Surgical Attending Physician', '666666666'),
25 (7, 'Bhavitha', 'Surgical Attending Physician', '777777777'),
26 (8, 'Naksh', 'MD Resident', '888888888'),
27 (9, 'Mahija', 'Attending Psychiatrist', '999999999');
28
29 -- Step 3: Create Department Table
30
31 CREATE TABLE department (
32     departmentid INT PRIMARY KEY,
33     name VARCHAR(100),
34     head INT
35 );
36

```

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help for the cur
or to toggle

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duplicates	Warnings
12	19:56:34	USE physician_department_db	0 row(s) affected	0	0
13	19:56:34	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR...	0 row(s) affected	0	0
14	19:56:34	INSERT INTO physician VALUES (1, 'Geethika', 'Staff Internist', '111111111), (2, 'S...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0	0
15	19:56:34	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARC...	0 row(s) affected	0	0

Object Info Session

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

Information

No object selected

SQL File 2*

Limit to 1000 rows

```

26 (8, 'Naksh', 'MD Resident', '888888888'),
27 (9, 'Mahija', 'Attending Psychiatrist', '999999999');
28
29 -- Step 3: Create Department Table
30
31 CREATE TABLE department (
32     departmentid INT PRIMARY KEY,
33     name VARCHAR(100),
34     head INT
35 );
36
37
38 -- Step 4: Insert Data into Department
39
40 INSERT INTO department VALUES
41 (1, 'General Medicine', 4),
42 (2, 'Surgery', 7),
43 (3, 'Psychiatry', 9);
44

```

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help for the current
or to toggle au

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duplicates	Warnings
19	19:56:57	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR...	0 row(s) affected	0	0
20	19:56:57	INSERT INTO physician VALUES (1, 'Geethika', 'Staff Internist', '111111111), (2, 'S...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0	0
21	19:56:57	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARC...	0 row(s) affected	0	0
22	19:56:57	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, P...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0	0

Object Info Session

Step 2 : JOIN

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File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 2' x

Limit to 1000 rows

SQLAdditions

Automatic context help for the current or to toggle auto

SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

Information

No object selected

43 (3, 'Psychiatry', 9);

44

45 -- Step 5: INNER JOIN

46

47 • SELECT d.name AS Department,

48 p.name AS Head_Physician

49 FROM department d

50 JOIN physician p

51 ON d.head = p.employeeid;

52

Result Grid

Department	Head_Physician
General Medicine	Jhansi
Surgery	Bhavitha
Psychiatry	Mahja

Result 1 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
27	19:57:26	INSERT INTO physician VALUES (1, 'Geethika', 'Staff Internist', '1111111111'); (2, ...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
28	19:57:26	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR...	0 row(s) affected
29	19:57:26	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
30	19:57:26	SELECT d.name AS Department, p.name AS Head_Physician FROM departm...	3 row(s) returned

Object Info Session

Step 3 : WHERE

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File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 2' x

Limit to 1000 rows

SQLAdditions

Automatic context help for the current or to toggle auto

SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

Information

No object selected

48 p.name AS Head_Physician

49 FROM department d

50 JOIN physician p

51 ON d.head = p.employeeid;

52

53 -- Step 6: WHERE Clause

54

55 • SELECT *

56 FROM physician

57 WHERE position = 'Surgical Attending Physician';

Result Grid

employeeid	name	position	ssn
3	Shailu	Surgical Attending Physician	33333333
6	Yamini	Surgical Attending Physician	66666666
7	Bhavitha	Surgical Attending Physician	77777777

Result 2 physician 3 x

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message
36	19:58:25	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR...	0 row(s) affected
37	19:58:25	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
38	19:58:25	SELECT d.name AS Department, p.name AS Head_Physician FROM departm...	3 row(s) returned
39	19:58:25	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIM...	3 row(s) returned

Object Info Session

Step 4 : LIKE

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 2*

SCHMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 2*

Limit to 1000 rows

```

56 FROM physician
57 WHERE position = 'Surgical Attending Physician';
58
59 -- Step 7: WHERE + LIKE
60
61 • SELECT *
62 FROM physician
63 WHERE name LIKE 'Harshika%';
64
65

```

Result Grid

employeeid	name	position	son
1	John	Dr.	Yes

Result 4 physician 5 physician 6 x

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message
46	19:58:42	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, '...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
47	19:58:42	SELECT d.name AS Department, p.name AS Head_Physician FROM departm...	3 row(s) returned
48	19:58:42	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIM...	3 row(s) returned
49	19:58:42	SELECT * FROM physician WHERE name LIKE 'Harshika%' LIMIT 0, 1000	0 row(s) returned

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Step 5 : COUNT

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File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 2*

SCHMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 2*

Limit to 1000 rows

```

59 -- Step 7: WHERE + LIKE
60
61 • SELECT *
62 FROM physician
63 WHERE name LIKE 'Harshika%';
64
65 -- Step 8: COUNT()
66
67 • SELECT COUNT(*) AS Total_Physicians
68 FROM physician;

```

Result Grid

Total_Physicians
9

Result 7 physician 8 physician 9 Result 10 x

Read Only

Output

Action Output

#	Time	Action	Message
57	19:59:33	SELECT d.name AS Department, p.name AS Head_Physician FROM departm...	3 row(s) returned
58	19:59:33	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIM...	3 row(s) returned
59	19:59:33	SELECT * FROM physician WHERE name LIKE 'Harshika%' LIMIT 0, 1000	0 row(s) returned
60	19:59:33	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned

mysql workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 2" x

Limit to 1000 rows

65 -- Step 8: COUNT()
66
67 • SELECT COUNT(*) AS Total_Physicians
68 FROM physician;
69
70 -- Step 9: COUNT DISTINCT
71
72 • SELECT COUNT(DISTINCT position) AS Different_Positions
73 FROM physician;
74

Result Grid

Filter Rows: Export: Wrap Cell Content:

Different_Positions
7

Administration Schemas

Information

No object selected

Object Info Session

Result 11 physician 12 physician 13 Result 14 Result 15 x

Output

Action Output

#	Time	Action	Message
69	19:59:51	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIM...	3 row(s) returned
70	19:59:51	SELECT * FROM physician WHERE name LIKE 'Harshika%' LIMIT 0, 1000	0 row(s) returned
71	19:59:51	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
72	19:59:51	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIM...	1 row(s) returned

Step 6 : GROUP BY

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 2" x

Limit to 1000 rows

71
72 • SELECT COUNT(DISTINCT position) AS Different_Positions
73 FROM physician;
74
75 -- Step 10: GROUP BY
76
77 • SELECT position,
78 COUNT(*) AS Total_Employees
79 FROM physician
80 GROUP BY position;

Result Grid

Filter Rows: Export: Wrap Cell Content:

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Administration Schemas

Information

No object selected

Object Info Session

Result 16 physician 17 physician 18 Result 19 Result 20 Result 21 x

Output

Action Output

#	Time	Action	Message
82	20:00:30	SELECT * FROM physician WHERE name LIKE 'Harshika%' LIMIT 0, 1000	0 row(s) returned
83	20:00:30	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
84	20:00:30	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIM...	1 row(s) returned
85	20:00:30	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned

30485 x

File Edit View Query Database Server Tools Scripting Help

SQL File 2*

Limit to 1000 rows

79 FROM physician
80
81
82 -- Step 11: GROUP BY + HAVING
83
84 • SELECT position,
85 COUNT(*) AS Total_Employees
86 FROM physician
87 GROUP BY position
88 HAVING COUNT(*) > 1

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Result Grid

position	Total_Employees
Surgical Attending Physician	3

Administration Schemas

Information

No object selected

Result 22 physician 23 physician 24 Result 25 Result 26 Result 27 Result 28 x

Read Only Context Help Snip

Output

#	Time	Action	Message
96	20:00:57	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
97	20:00:57	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT...	1 row(s) returned
98	20:00:57	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned
99	20:00:57	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	1 row(s) returned

Object Info Session

Step 7: ORDER BY

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 2*

Limit to 1000 rows

86 FROM physician
87 GROUP BY position
88 HAVING COUNT(*) > 1;
89
90 -- Step 12: ORDER BY ASC
91
92 • SELECT *
93 FROM physician
94 ORDER BY name ASC;
95

Administration Schemas

Information

No object selected

Result Grid

employeeid	name	position	ssn
5	Aishu	Head Chief of Medicine	555555555
7	Bhavitha	Surgical Attending Physician	555555555
1	Geethika	Staff Internist	111111111
4	Jhansi	Senior Attending Physician	444444444
9	Mahija	Attending Psychiatrist	999999999
8	Naksh	MD Resident	888888888
3	Shaili	Surgical Attending Physician	333333333

Result 29 physician 30 physician 31 Result 32 Result 33 Result 34 Result 35 physician 36 x Apply Conte

Output

Action Output

#	Time	Action	Message
✓ 111	20:01:29	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIM...	1 row(s) returned
✓ 112	20:01:29	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned
✓ 113	20:01:29	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	1 row(s) returned
✓ 114	20:01:29	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned

Object Info Session

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 2*

Limit to 1000 rows

Automatic context help
Use the toolbar to help for the current or to toggle auto

SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

No object selected

```

91
92 • SELECT *
93 FROM physician
94 ORDER BY name ASC;
95
96 -- Step 13: ORDER BY DESC
97
98 • SELECT *
99 FROM physician
100 ORDER BY employeeid DESC;

```

employeeid	name	position	ssn
9	Mahija	Attending Psychiatrist	999999999
8	Naksh	MD Resident	888888888
7	Bhavitha	Surgical Attending Physician	777777777
6	Yamini	Surgical Attending Physician	666666666
5	Aishu	Head Chief of Medicine	555555555
4	Jhansi	Senior Attending Physician	444444444
3	Shaili	Surinical Attending Physician	333333333

physician 38 physician 39 Result 40 Result 41 Result 42 Result 43 physician 44 physician Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message
127	20:01:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned
128	20:01:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	1 row(s) returned
129	20:01:50	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
130	20:01:50	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned

Object Info Session

Step 8 : IN

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SQL File 2*

Limit to 1000 rows

Author Use help or

SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

Administration Schemas

No object selected

```

100 ORDER BY employeeid DESC;
101
102 -- Step 14: WHERE + IN
103
104 • SELECT *
105 FROM physician
106 WHERE employeeid IN (
107     SELECT head
108     FROM department
109 );

```

employeeid	name	position	ssn
9	Mahija	Attending Psychiatrist	999999999
8	Naksh	MD Resident	888888888
7	Bhavitha	Surgical Attending Physician	777777777
6	Yamini	Surgical Attending Physician	666666666
5	Aishu	Head Chief of Medicine	555555555
4	Jhansi	Senior Attending Physician	444444444
3	Shaili	Surinical Attending Physician	333333333

physician 38 physician 39 Result 40 Result 41 Result 42 Result 43 physician 44 physician Apply Context Help

Output

Action Output

#	Time	Action	Message
127	20:01:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned
128	20:01:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	1 row(s) returned
129	20:01:50	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
130	20:01:50	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned

Object Info Session

Step 9 : NOT IN

The screenshot shows a database management tool interface. On the left, the 'SCHEMAS' pane lists several databases: aggregate_bank_db, bank_management_db, bookflow_db, joins_operations_db, std_marks, and sys. The main editor displays a SQL query for Step 15: NOT IN. The query selects all records from the 'physician' table where the 'employeeid' is not in the 'head' column of the 'department' table. Below the query, the 'Result Grid' shows a table with 8 rows of physician data. The 'Output' pane at the bottom shows the execution log, indicating that the query returned 6 rows.

```
110
111 -- Step 15: NOT IN
112
113 • SELECT *
114   FROM physician
115  WHERE employeeid NOT IN (
116      SELECT head
117      FROM department
118  );
119
```

employeeid	name	position	ssn
1	Geethika	Staff Internist	111111111
2	Siri	Attending Physician	222222222
3	Shailu	Surgical Attending Physician	333333333
5	Aishu	Head Chief of Medicine	555555555
6	Yamini	Surgical Attending Physician	666666666
8	Naksh	MD Resident	888888888
NULL	NULL	NULL	NULL

#	Time	Action	Message
✓ 145	20:06:35	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 146	20:06:35	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 147	20:06:35	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM d...	3 row(s) returned
✓ 148	20:06:35	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FR...	6 row(s) returned

Step 10 : GROUP BY + ORDER BY

The screenshot displays the SQL Enterprise Manager interface. The left pane shows the 'SCHEMAS' tree with databases like 'aggregate_bank_db', 'bank_management_db', 'bookflow_db', 'joins_operations_db', 'std_marks', and 'sys'. The main query editor shows the following SQL code:

```
119
120 -- Step 16: GROUP BY + ORDER BY
121
122 • SELECT position,
123       COUNT(*) AS Total_Employees
124 FROM physician
125 GROUP BY position
126 ORDER BY Total_Employees DESC;
127
128
```

The 'Result Grid' shows the following data:

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

The 'Output' pane shows the 'Action Output' table with the following rows:

#	Time	Action	Message
✓ 164	20:06:56	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 165	20:06:56	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM d...	3 row(s) returned
✓ 166	20:06:56	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FR...	6 row(s) returned
✓ 167	20:06:56	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned

Step 11: LIKE

The screenshot shows a SQL IDE interface with a query editor and a result grid. The query editor contains the following SQL code:

```
124 FROM physician
125 GROUP BY position
126 ORDER BY Total_Employees DESC;
127
128 -- Step 17: WHERE + LIKE
129
130 • SELECT *
131 FROM physician
132 WHERE position LIKE '%Attending%';
133
```

The result grid displays the following data:

employeeid	name	position	ssn
2	Siri	Attending Physician	222222222
3	Shailu	Surgical Attending Physician	333333333
4	Jhansi	Senior Attending Physician	444444444
6	Yamini	Surgical Attending Physician	666666666
7	Bhavitha	Surgical Attending Physician	777777777
9	Mahija	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

The output pane shows the following messages:

#	Time	Action	Message
✓ 184	20:07:33	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM d...	3 row(s) returned
✓ 185	20:07:33	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FR...	6 row(s) returned
✓ 186	20:07:33	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned
✓ 187	20:07:33	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned

Step 12: JOIN + WHERE

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File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

SQL File 2*

```

131 FROM physician
132 WHERE position LIKE '%Attending%';
133
134 -- Step 18: JOIN + WHERE
135
136 • SELECT p.name
137 FROM physician p
138 JOIN department d
139 ON p.employeeid = d.head
140 WHERE d.name = 'Surgery';

```

Result Grid

employeeid	name	position	ssn
2	Siri	Attending Physician	222222222
3	Shalu	Surgical Attending Physician	333333333
4	Jhansi	Senior Attending Physician	444444444
6	Yamini	Surgical Attending Physician	666666666
7	Bhavitha	Surgical Attending Physician	777777777
9	Mahija	Attending Psychiatrist	999999999

Output

#	Time	Action	Message
184	20:07:33	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM d...	3 row(s) returned
185	20:07:33	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FR...	6 row(s) returned
186	20:07:33	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned
187	20:07:33	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned

Object Info Session

Step 13: JOIN+ORDER BY

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

Filter objects

- aggregate_bank_db
- bank_management_db
- bookflow_db
- joins_operations_db
- std_marks
- sys

SQL File 2*

```

140 WHERE d.name = 'Surgery';
141
142 -- Step 19: JOIN + ORDER BY
143
144 • SELECT d.name AS Department,
145        p.name AS Head_Physician
146 FROM department d
147 JOIN physician p
148 ON d.head = p.employeeid
149 ORDER BY d.name;

```

Result Grid

Department	Head_Physician
General Medicine	Jhansi
Psychiatry	Mahija
Surgery	Bhavitha

Output

#	Time	Action	Message
206	20:08:13	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP B...	7 row(s) returned
207	20:08:13	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned
208	20:08:13	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head...	1 row(s) returned
209	20:08:13	SELECT d.name AS Department, p.name AS Head_Physician FROM departm...	3 row(s) returned

Object Info Session

Step 14: NOT LIKE

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 2" x

Limit to 1000 rows

```

146 FROM department d
147 JOIN physician p
148 ON d.head = p.employeeid
149 ORDER BY d.name;
150
151 -- Step 20: WHERE + NOT LIKE
152
153 • SELECT *
154 FROM physician
155 WHERE position NOT LIKE '%Surgical%';

```

Result Grid

employeeid	name	position	ssn
1	Geethika	Staff Internist	111111111
2	Siri	Attending Physician	222222222
4	Jhansi	Senior Attending Physician	444444444
5	Aishu	Head Chief of Medicine	555555555
8	Naksh	MD Resident	888888888
9	Mahija	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Output

Action Output

#	Time	Action	Message
229	20:08:47	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned
230	20:08:47	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head...	1 row(s) returned
231	20:08:47	SELECT d.name AS Department, p.name AS Head_Physician FROM departm...	3 row(s) returned
232	20:08:47	SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000	6 row(s) returned

Object Info Session

Step 15: COUNT+HAVING

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 2" x

Limit to 1000 rows

```

154 FROM physician
155 WHERE position NOT LIKE '%Surgical%';
156
157 -- Step 21: COUNT + HAVING
158
159 • SELECT position,
160 COUNT(*) AS Total
161 FROM physician
162 GROUP BY position
163 HAVING COUNT(*) >= 2;

```

Result Grid

position	Total
Surgical Attending Physician	3

Output

Action Output

#	Time	Action	Message
253	20:09:14	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head...	1 row(s) returned
254	20:09:14	SELECT d.name AS Department, p.name AS Head_Physician FROM departm...	3 row(s) returned
255	20:09:14	SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000	6 row(s) returned
256	20:09:14	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position H...	1 row(s) returned

Object Info Session

Step 16: EXISTS

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the 'SCHEMAS' tree with a search filter 'Filter objects'. The main pane shows a query in 'SQL File 2*' with the following code:

```
165
166 -- Step 22: EXISTS
167
168 • SELECT *
169 FROM department d
170 WHERE EXISTS (
171     SELECT 1
172     FROM physician p
173     WHERE p.employeeid = d.head
174 );
```

Below the query editor, the 'Result Grid' shows the results of the query. The grid has columns 'departmentid', 'name', and 'head'. The results are:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

The bottom pane shows the 'Output' window with the 'Action Output' tab selected. It displays a list of actions and their results:

#	Time	Action	Message
278	20:09:55	SELECT d.name AS Department, p.name AS Head_Physician FROM department d, physician p WHERE p.employeeid = d.head	3 row(s) returned
279	20:09:55	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
280	20:09:55	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) > 1	1 row(s) returned
281	20:09:55	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid = d.head)	3 row(s) returned

Step 17: MAX

SQL File 2* x

Limit to 1000 rows

```

170 WHERE EXISTS (
171     SELECT 1
172     FROM physician p
173     WHERE p.employeeid = d.head
174 );
175
176 -- Step 23: MAX()
177
178 • SELECT MAX(employeeid) AS Highest_Employee_ID
179 FROM physician;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Highest_Employee_ID
▶	9

Alt 159 physician 160 Result 161 Result 162 physician 163 Result 164 department 165 Result

Output

Action Output

#	Time	Action	Message
✓ 304	20:10:12	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓ 305	20:10:12	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position H...	1 row(s) returned
✓ 306	20:10:12	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physicia...	3 row(s) returned
✓ 307	20:10:12	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 10...	1 row(s) returned

Step 18: MIN

30485 x

FileEditViewQueryDatabaseServerToolsScriptingHelp

SQL File 2" x

Limit to 1000 rows

SCHEMAS

Filter objects

▶ aggregate_bank_db

▶ bank_management_db

▶ bookflow_db

▶ joins_operations_db

▶ std_marks

▶ sys

SQL Editor

175

176 -- Step 23: MAX()

177

178 • SELECT MAX(employeeid) AS Highest_Employee_ID

179 FROM physician;

180

181 -- Step 24: MIN()

182

183 • SELECT MIN(employeeid) AS Lowest_Employee_ID

184 FROM physician;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Form Editor

Lowest_Employee_ID

1

Administration

Schemas

Information

No object selected

Object Info

Session

physician 179

Result 180

Result 181

physician 182

Result 183

department 184

Result 185

Result 186

Read Only

Output

Action Output

#	Time	Action	Message
✓ 331	20:10:37	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position H...	1 row(s) returned
✓ 332	20:10:37	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physicia...	3 row(s) returned
✓ 333	20:10:37	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 10...	1 row(s) returned
✓ 334	20:10:37	SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned